

Detailed Balance for Effective One-Dimensional Reaction Kernels

1 Reviewer comment and proposed response

Reviewer comment

A related question concerns the use of detailed-balance emission rates together with effective sink-strength reaction kernels. While Eq. (70) is standard for conventional second-order absorption reactions, the manuscript also relies on reaction kernels derived from 1D-mobile defect/fixed-sink formulations. It is unclear whether the same detailed-balance arguments remain applicable in such third-order reactions. At the very least, the assumptions and limitations underlying this approximation should be discussed.

Proposed response

We thank the reviewer for raising this important point. We agree that the detailed-balance relation conventionally used for a binary absorption reaction cannot be transferred without qualification to an effective, concentration-dependent sink-strength kernel.

The pure-1D absorption law is not interpreted here as an elementary three-body reaction. It results from averaging the first-passage motion of a 1D-mobile cluster over the entire fixed-sink population. Its apparent third-order dependence arises because one factor of sink concentration specifies the family responsible for capture, whereas the total macroscopic cross section determines the mean distance between intersections of the glide trajectory and all sinks [4, 5, 6].

Detailed balance applies rigorously to the underlying reversible elementary transition $A + B \rightleftharpoons C$, rather than automatically to its coarse-grained multisink closure. At equilibrium,

$$K_{AB}^{\text{eff}}(c^{\text{eq}})c_A^{\text{eq}}c_B^{\text{eq}} = \alpha_C^{\text{eff}}(c^{\text{eq}})c_C^{\text{eq}}.$$

Thus, if the concentration-dependent 1D absorption kernel is retained in the equilibrium relation, the thermodynamically conjugate effective emission coefficient must contain the same environmental dependence. Combining an instantaneous 1D sink-strength kernel with a concentration-independent emission rate obtained from the conventional binary formula is not an exact detailed-balance construction.

In the present model, the detailed-balance emission expression is therefore regarded as a dilute-medium, local-equilibrium approximation. Emission is assumed to be governed by the binding free energy of the emitting cluster, whereas the multisink transport correction affects capture without changing the intrinsic detachment event. The revised manuscript states this assumption and its limitations. The approximation may lose thermodynamic consistency when sink populations change substantially, when 1D transport produces strong spatial correlations, or when the system approaches thermal equilibrium.

If emission is demonstrably small in the simulated regime, one may add:

Because the calculations address a strongly driven irradiation regime in which production and absorption fluxes are much greater than the relevant thermally activated emission fluxes, this approximation does not control the predicted microstructure. Nevertheless, its thermodynamic status and range of validity are now stated explicitly.

2 Detailed balance for a general reaction

Consider

$$\sum_i \nu_{ir}^- X_i \rightleftharpoons \sum_i \nu_{ir}^+ X_i, \quad \Delta \nu_{ir} = \nu_{ir}^+ - \nu_{ir}^-. \quad (1)$$

For ideal mass-action kinetics,

$$J_r^+ = k_r^+ \prod_i a_i^{\nu_{ir}^-}, \quad J_r^- = k_r^- \prod_i a_i^{\nu_{ir}^+}, \quad (2)$$

where a_i is a dimensionless activity. Microscopic reversibility requires $J_r^{+,eq} = J_r^{-,eq}$ for every reversible channel. Consequently,

$$\boxed{\frac{k_r^+}{k_r^-} = \prod_i (a_i^{eq})^{\Delta \nu_{ir}} \equiv K_r^{eq}}, \quad \boxed{k_r^- = \frac{k_r^+}{K_r^{eq}}}. \quad (3)$$

When dimensional concentrations replace activities, powers of the standard concentration must be retained because forward and reverse constants generally have different dimensions [9, 10].

3 Conventional binary absorption and emission

For $X_n + X_1 \rightleftharpoons X_{n+1}$, write

$$J_n^+ = \beta_n c_n c_1, \quad J_{n+1}^- = \alpha_{n+1} c_{n+1}.$$

Detailed balance gives

$$\boxed{\alpha_{n+1} = \beta_n \frac{c_n^{eq} c_1^{eq}}{c_{n+1}^{eq}}}. \quad (4)$$

If

$$\frac{c_{n+1}^{eq}}{c_n^{eq} c_1^{eq}} = g_{n+1} \exp \left(\frac{\Delta G_{n+1}^b}{k_B T} \right),$$

then

$$\boxed{\alpha_{n+1} = \frac{\beta_n}{g_{n+1}} \exp \left(-\frac{\Delta G_{n+1}^b}{k_B T} \right)}. \quad (5)$$

Here g_{n+1} contains standard-state and configurational-degeneracy factors. Replacing binding free energy by binding energy assumes that the relevant entropy is negligible or included in the prefactor.

4 Pure-1D absorption as an effective higher-order law

Let 1D-mobile species A be captured by fixed clusters of family B . Define

$$\Sigma_A = \sum_j \sigma_{Aj} N_j. \quad (6)$$

The family-resolved sink strength and per- A -cluster frequency are

$$k_{(1),A \rightarrow B}^2 = 6\sigma_{AB} N_B \Sigma_A, \quad \mathcal{K}_{A \rightarrow B}^{(1)} = 6D_A \sigma_{AB} N_B \sum_j \sigma_{Aj} N_j. \quad (7)$$

The rate density is

$$\boxed{J_{A+B}^+ = 6D_A \sigma_{AB} N_A N_B \sum_j \sigma_{Aj} N_j} = \sum_j 6D_A \sigma_{AB} \sigma_{Aj} N_A N_B N_j. \quad (8)$$

This resembles $A + B + X_j \rightarrow C + X_j$, but X_j is not normally a third defect in the local capture event. Its population determines the ensemble spacing between intersections of the 1D path and sinks. The third factor is environmental and geometrical, generated by coarse graining [4, 5, 8].

For one sink family,

$$\boxed{J_{A+B}^+ = 6D_A \sigma_{AB}^2 N_A N_B^2}. \quad (9)$$

With $\sigma_{AB} = \pi R_{AB}^2$,

$$J_{A+B}^+ = 6\pi^2 D_A R_{AB}^4 N_A N_B^2, \quad (10)$$

displaying the effective third-order and R^4 dependences.

5 Detailed balance for the effective 1D closure

For $A + B \rightleftharpoons C$, write

$$J^+ = K_{AB}^{\text{eff}}(\mathbf{N}) N_A N_B, \quad K_{AB}^{\text{eff}}(\mathbf{N}) = 6D_A \sigma_{AB} \Sigma_A, \quad (11)$$

and $J^- = \alpha_C^{\text{eff}}(\mathbf{N}) N_C$. At equilibrium,

$$\boxed{\alpha_C^{\text{eff}}(\mathbf{N}^{\text{eq}}) = K_{AB}^{\text{eff}}(\mathbf{N}^{\text{eq}}) \frac{N_A^{\text{eq}} N_B^{\text{eq}}}{N_C^{\text{eq}}}}. \quad (12)$$

Defining

$$K_{\text{th}} = \frac{N_C^{\text{eq}}}{N_A^{\text{eq}} N_B^{\text{eq}}},$$

gives

$$\boxed{\alpha_C^{\text{eff}}(\mathbf{N}^{\text{eq}}) = \frac{K_{AB}^{\text{eff}}(\mathbf{N}^{\text{eq}})}{K_{\text{th}}}}. \quad (13)$$

This guarantees zero net flux at that equilibrium; it does not define an intrinsic detachment frequency away from that state.

5.1 Instantaneously consistent coarse-grained closure

One can impose

$$\alpha_C^{\text{eff}}(\mathbf{N}) = K_{AB}^{\text{eff}}(\mathbf{N})/K_{\text{th}}$$

at every instant. Then

$$J_{\text{net}} = K_{AB}^{\text{eff}}(\mathbf{N}) \left(N_A N_B - \frac{N_C}{K_{\text{th}}} \right). \quad (14)$$

This preserves the equilibrium manifold, but makes the effective emission frequency depend on the external sink population through Σ_A . It is a consistent coarse-grained reverse flux, not the intrinsic frequency at which C emits A .

6 Formulation for genuine higher-order reactions

For

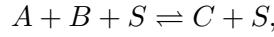
$$\sum_i \alpha_i X_i \rightleftharpoons \sum_i \beta_i X_i,$$

the generalized mass-action flux is

$$J_{\text{net}} = k^+ \prod_i a_i^{\alpha_i} - k^- \prod_i a_i^{\beta_i}, \quad \frac{k^+}{k^-} = \prod_i (a_i^{\text{eq}})^{\beta_i - \alpha_i}. \quad (15)$$

Detailed balance therefore applies to any molecularity, provided the forward and reverse laws represent the same elementary channel and use consistent activities.

For the formal catalytic reaction



$$J^+ = k^+ a_A a_B a_S, \quad J^- = k^- a_C a_S.$$

The catalyst cancels from the equilibrium constant,

$$\frac{k^+}{k^-} = \frac{a_C^{\text{eq}}}{a_A^{\text{eq}} a_B^{\text{eq}}}, \quad (16)$$

but multiplies both nonequilibrium fluxes:

$$J_{\text{net}} = a_S (k^+ a_A a_B - k^- a_C).$$

Thus a formally reversible multisink closure is

$$J_{\text{net}} = 6D_A \sigma_{AB} \Sigma_A \left(N_A N_B - \frac{N_C}{K_{\text{th}}} \right). \quad (17)$$

This is a reversible coarse-grained equation, but its reverse term is not an intrinsic evaporation rate.

7 Recommended physical formulation

The cleanest treatment separates intrinsic detachment from transport and recapture:

$$\alpha_C^{\text{int}} = \nu_C \exp \left(-\frac{\Delta G_C^\ddagger}{k_B T} \right). \quad (18)$$

If

$$\Delta G_C^\ddagger \simeq \Delta G_A^m + \Delta G_C^b,$$

then

$$\alpha_C^{\text{int}} \simeq \nu_C \exp \left[-\frac{\Delta G_A^m + \Delta G_C^b}{k_B T} \right]. \quad (19)$$

The intrinsic event should not depend on unrelated external sinks. After emission, the released defect undergoes 1D, 1DR, or 3D transport and possible recapture, described separately by the appropriate transport kernel.

8 Modeling alternatives and limitations

1. **Mechanistic formulation (recommended):** compute intrinsic emission from the detachment barrier and use the 1D or 1DR kernel only for subsequent capture and recapture.
2. **Reversible coarse-grained formulation:** multiply the reverse flux by the same concentration-dependent transport factor as the forward flux, as in Eq. (14). This preserves equilibrium but makes the reverse coefficient an effective environmental rate.
3. **Dilute nonequilibrium approximation:** retain conventional binary detailed-balance emission while using the effective 1D absorption kernel. This may be adequate when emission is demonstrably negligible, but it is not an exact detailed-balance construction.

The third approximation may fail when sink populations evolve strongly, spatial correlations invalidate the homogeneous closure, rotation or trapping changes migration dimensionality, emission and recapture occur on comparable time scales, or the system approaches thermal equilibrium.

9 Suggested manuscript insertion

Detailed balance for concentration-dependent kernels. The detailed-balance relation conventionally used for $X_n + X_1 \rightleftharpoons X_{n+1}$ is exact when the forward and reverse terms represent the same pair of elementary transitions. The sink-strength expression for pure 1D migration has a different status:

$$K_{n,1}^{\text{eff}}(\mathbf{N}) = 6D_1\sigma_{1n} \sum_j \sigma_{1j} N_j, \quad J_{n,1}^+ = K_{n,1}^{\text{eff}}(\mathbf{N}) N_n N_1.$$

Its apparent higher-order dependence is not an elementary three-defect collision. The additional factor describes how the sink population determines the mean distance between intersections of a 1D trajectory and absorbing sinks [4, 5, 6].

At equilibrium, a coarse-grained reverse coefficient must satisfy

$$\alpha_{n+1}^{\text{eff}}(\mathbf{N}^{\text{eq}}) = K_{n,1}^{\text{eff}}(\mathbf{N}^{\text{eq}}) \frac{N_n^{\text{eq}} N_1^{\text{eq}}}{N_{n+1}^{\text{eq}}}.$$

Thus, applying the conventional binary detailed-balance expression with an instantaneous concentration-dependent 1D kernel is not an exact microscopic construction unless the reverse term contains the same environmental dependence. Here thermal emission is treated as an intrinsic unimolecular detachment process governed by cluster binding free energy, whereas the 1D sink-strength factor describes subsequent transport-controlled capture. Use of the binary expression with an effective 1D kernel is therefore a dilute-medium, local-equilibrium approximation. It may become inaccurate when sink populations evolve strongly, spatial correlations are important, or the system approaches thermal equilibrium.

References

References

- [1] M. von Smoluchowski, “Versuch einer mathematischen Theorie der Koagulationskinetik kolloider Lösungen,” *Z. Phys. Chem.* **92** (1917) 129–168.
- [2] T. R. Waite, “Theoretical treatment of the kinetics of diffusion-limited reactions,” *Phys. Rev.* **107** (1957) 463–470, [doi:10.1103/PhysRev.107.463](https://doi.org/10.1103/PhysRev.107.463).
- [3] U. Gösele and A. Seeger, “Theory of bimolecular reaction rates limited by anisotropic diffusion,” *Philos. Mag.* **34** (1976) 177–193, [doi:10.1080/14786437608223764](https://doi.org/10.1080/14786437608223764).
- [4] V. A. Borodin, “Rate theory for one-dimensional diffusion,” *Physica A* **260** (1998) 467–478, [doi:10.1016/S0378-4371\(98\)00338-0](https://doi.org/10.1016/S0378-4371(98)00338-0).
- [5] A. V. Barashev, S. I. Golubov, and H. Trinkaus, “Reaction kinetics of glissile interstitial clusters in a crystal containing voids and dislocations,” *Philos. Mag. A* **81** (2001) 2515–2532, [doi:10.1080/01418610108217159](https://doi.org/10.1080/01418610108217159).
- [6] H. Trinkaus, H. L. Heinisch, A. V. Barashev, S. I. Golubov, and B. N. Singh, “1D to 3D diffusion-reaction kinetics of defects in crystals,” *Phys. Rev. B* **66** (2002) 060105(R), [doi:10.1103/PhysRevB.66.060105](https://doi.org/10.1103/PhysRevB.66.060105).
- [7] S. Huang and J. Marian, “Rates of diffusion-controlled reactions for one-dimensionally moving species in 3D space,” *Philos. Mag.* **99** (2019) 2562–2583, [doi:10.1080/14786435.2019.1632930](https://doi.org/10.1080/14786435.2019.1632930).
- [8] G. Adjanor, “Complete characterization of sink strengths for mutually 1D-mobile defect clusters: Extension to diffusion-anisotropy analog cases,” *J. Nucl. Mater.* **570** (2022) 153970, [doi:10.1016/j.jnucmat.2022.153970](https://doi.org/10.1016/j.jnucmat.2022.153970).
- [9] T. L. Hill, *Free Energy Transduction and Biochemical Cycle Kinetics*, Springer-Verlag, New York, 1989.
- [10] H. Qian, “Statistical chemical thermodynamics and energetic behavior of molecular systems: From equilibrium to nonequilibrium,” *J. Phys. Chem. B* **126** (2022).